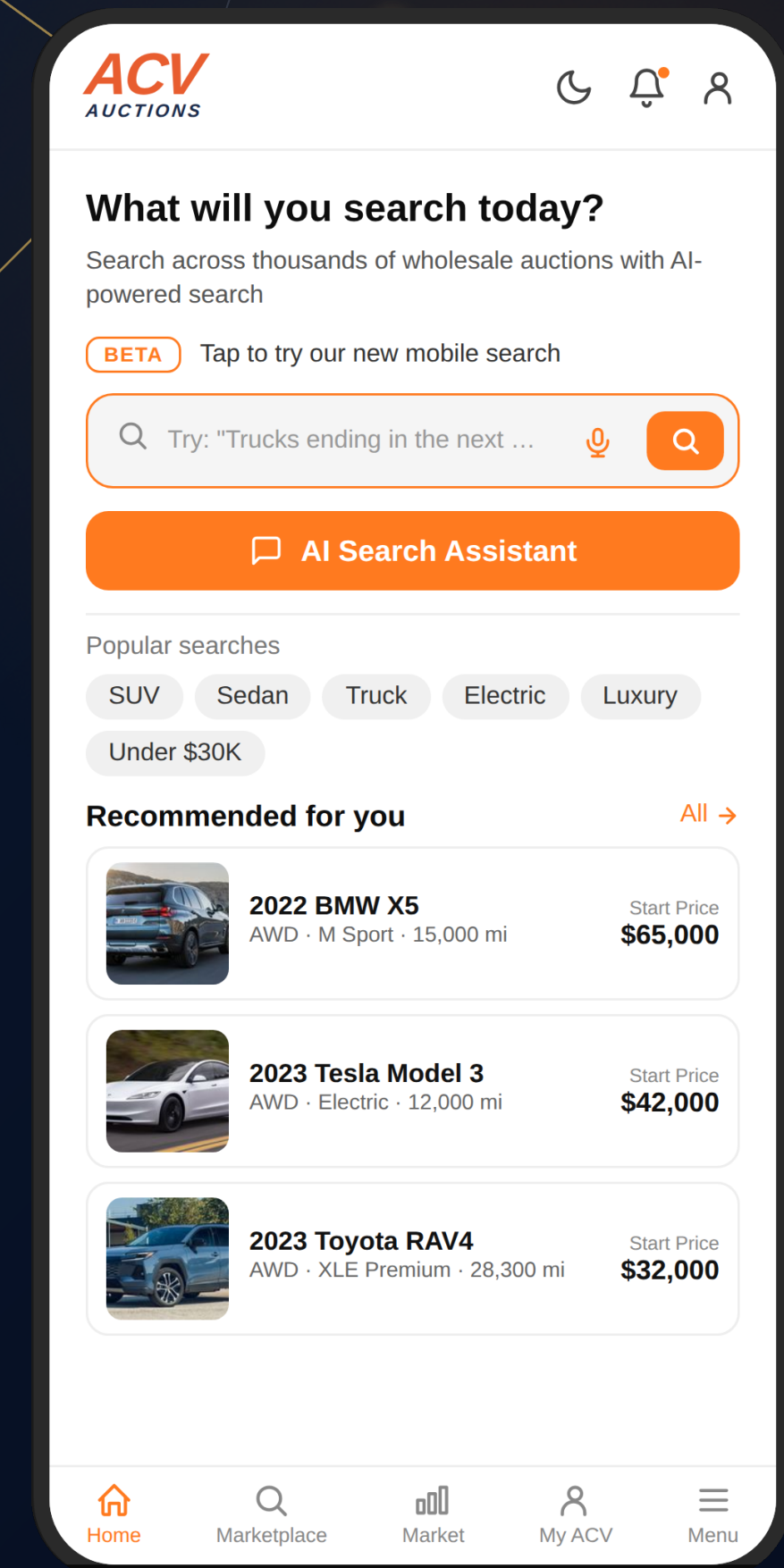


## PRODUCT DESIGN CASE STUDY

# ACV EASE

*AI-powered search & discovery for wholesale vehicle auctions.*

Margaux Kan · Senior Product Design Manager



## TITLE & OUTCOME

# Reimagining how dealers find their next car

ACV EASE is a mobile-first search experience for ACV's wholesale auction marketplace. Instead of stacking rigid filters, buyers ask for what they want in plain language, by voice or text — and an AI layer surfaces the strongest opportunities, explains why they matter, and guides each buyer from search to bid.

## MY ROLE

**Lead Product Designer**

Research · IxD · UI · Prototyping

## TIMELINE

**14 weeks, 2025**

## PLATFORM

**iOS & Android — ACV mobile app**

## TEAM

**1 PM · 4 Engineers · 2 Data Scientists**

## STRATEGIC PROBLEM & RISK

# Finding the right car was slow, manual, and overwhelming

Buyers run a high-volume business from their phones — between appraisals, on the lot, on the move. The old search made them work for it.

### Filters, not language

Buyers had to translate what they wanted into rigid drop-downs and checkboxes, often missing inventory that didn't match the exact filter.

### Thousands of live auctions

Relevant vehicles were scattered across constantly-changing auctions, hard to scan on a small screen with seconds to spare.

### Good deals stayed buried

Nothing told a buyer which listing was actually a strong opportunity, so the best cars were easy to scroll right past.

## MY ROLE & LEADERSHIP

# Leading search from a strategic bet to launch

Lead product designer: research, interaction design, UI, and prototyping across the search-to-bid journey.

### 01 WHAT I LED

- Framed AI-powered search as a growth bet, not a UI refresh.
- Defined the vision for conversational, explainable discovery.
- Aligned Product, Engineering, and Data Science around buyer intent.
- Owned the search experience from discovery through beta launch.

### 02 HOW I LED

- Championed a mobile-first experience built for one hand on the lot.
- Partnered with Data Science on ranking and plain-language reasons.
- Led 3 design reviews; balanced speed against buyer trust.
- Used field evidence to resolve trade-offs across teams.

## DESIGN PROCESS

# A structured path from intent to confident bid

Five phases — pushing design upstream to shape the search strategy, not just the UI.

01

### Understand

Studied how dealers search between appraisals and on the move.

02

### Frame

Mapped the search-to-bid journey and a buyer-intent model.

03

### Explore

Co-designed conversational and voice search patterns.

04

### Validate

Prototyped & tested natural-language and voice search with dealers.

05

### Deliver

Shipped a beta with Product, Engineering & Data Science.

## LEADERSHIP LENS

*Embedded design early to influence the ranking and intent strategy, not just the screens.*



## RESEARCH

# Grounding AI search in how buyers actually search

Research led by me, with Product and Data Science embedded as partners.

### METHODS

- Interviews & ride-alongs with 10 dealers across the U.S.
- Observed live search sessions on the legacy filter experience.
- Audited the old search; usability-tested NL & voice prototypes.

### HOW RESEARCH WAS RUN

- Research plan led and owned by me.
- PMs embedded as co-researchers; Data Science on intent signals.
- Findings shared across Product, Engineering & Data Science.

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**GOAL** *Learn where filter search broke down, and what a “good result” means to a buyer with seconds to decide.*

USER JOURNEY

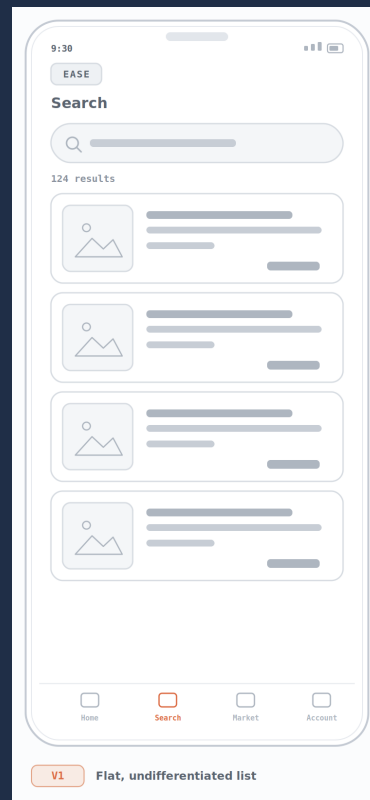
From first need to confident bid

|               | DISCOVER                                          |                                               | EVALUATE                                       | COMMIT                                            |                                                   |
|---------------|---------------------------------------------------|-----------------------------------------------|------------------------------------------------|---------------------------------------------------|---------------------------------------------------|
| STAGE         | 1. Spot a need                                    | 2. Search                                     | 3. Sift results                                | 4. Size up the car                                | 5. Decide & bid                                   |
| DOING         | On the lot                                        | Voice / text                                  | Results list                                   | Vehicle page                                      | Bid + assistant                                   |
| FEELING       | <div>😊 Hopeful</div>                              | <div>😞 Friction</div>                         | <div>😓 Overwhelmed</div>                       | <div>😟 Unsure</div>                               | <div>😰 Anxious</div>                              |
| PAIN POINT    | <i>Restocking is reactive, time-boxed.</i>        | <i>Wants don't fit rigid dropdowns.</i>       | <i>Strong cars buried in the noise.</i>        | <i>No read on whether the deal is good.</i>       | <i>Commits on gut, low confidence.</i>            |
| EASE RESPONSE | <div>Meets buyers on mobile, in the moment.</div> | <div>Plain-language &amp; voice search.</div> | <div>Surfaces one ranked top pick + why.</div> | <div>AI summary, bid score, success signal.</div> | <div>Confident bid; assistant one tap away.</div> |

## INSIGHTS & ITERATIONS

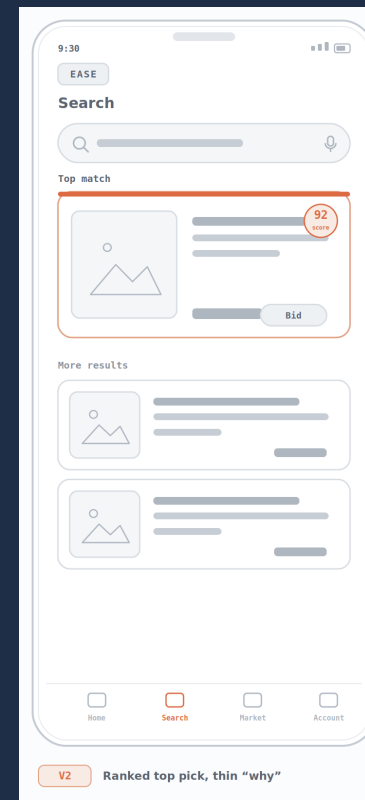
# What we heard, and how the design evolved

Three insights reshaped the experience from a list into a recommendation.



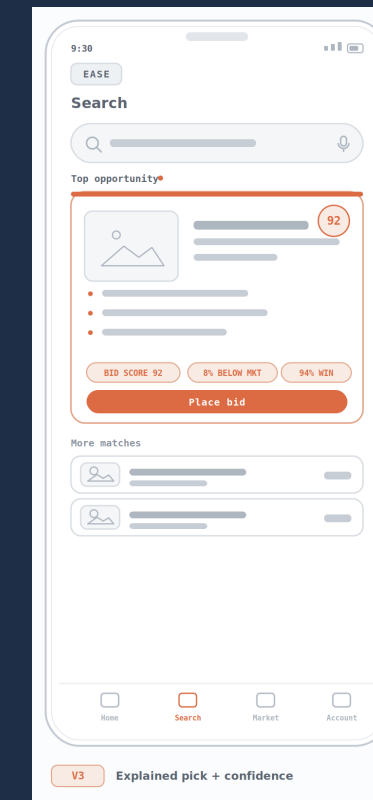
### 01 V1 — flat list

Buyers think in plain language,  
not filter taxonomies.



### 02 V2 — ranked pick

One strong, explained pick beats  
a long list of options.



### 03 V3 — explained + confidence

Trust comes from the why: reasons  
plus a confidence signal.

**KEY LEARNING** *Leading with one explained opportunity, plus plain-language and voice input, cut search friction without hiding inventory.*



# From filtering a catalog to surfacing the best opportunity

## PRIMARY PERSONA

### High-Volume Dealer

- Mobile-first; buys between appraisals and on the lot.
- Optimizes for speed, margin, and the next profitable car.
- Low tolerance for friction; high need for trust.

## STRATEGIC REFRAME

**We shifted from helping buyers filter inventory to recommending the strongest opportunity — and explaining why it's worth a bid.**

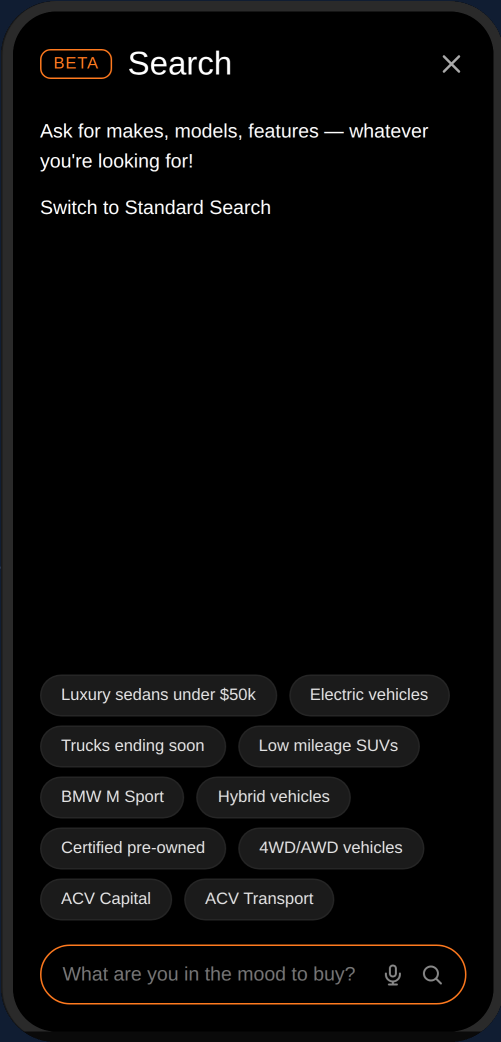
*“EASE surfaces the one car I'd have missed. It's like having a co-buyer in my pocket.”*

Jason M., Independent Dealer — Texas

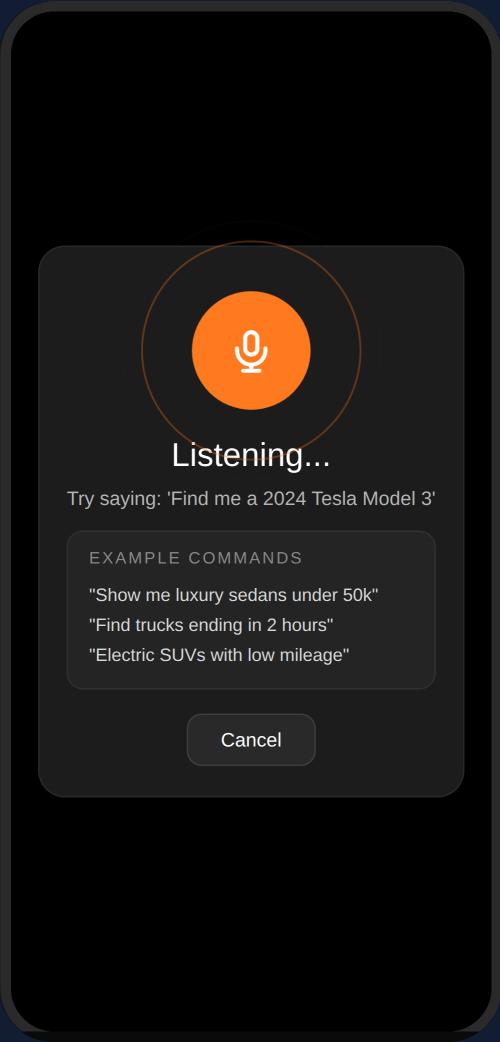
THE SOLUTION

# One intelligent flow, end to end

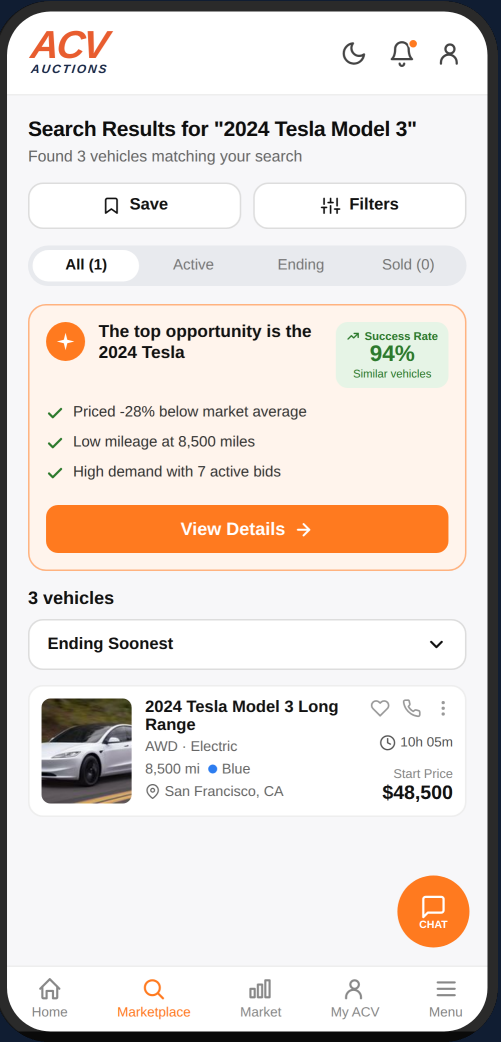
From a spoken request to a confident bid, the experience stays in one continuous, guided thread.



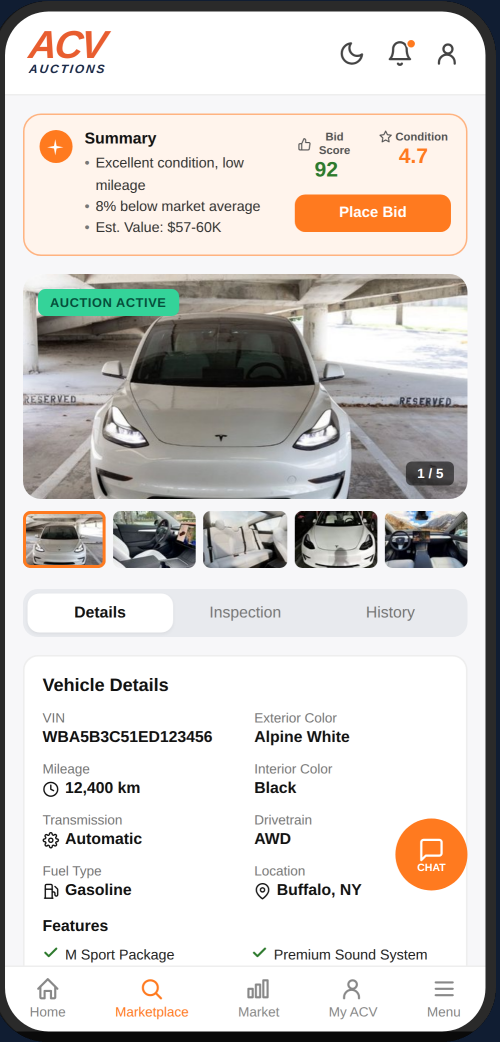
1Ask



2Understand



3Surface



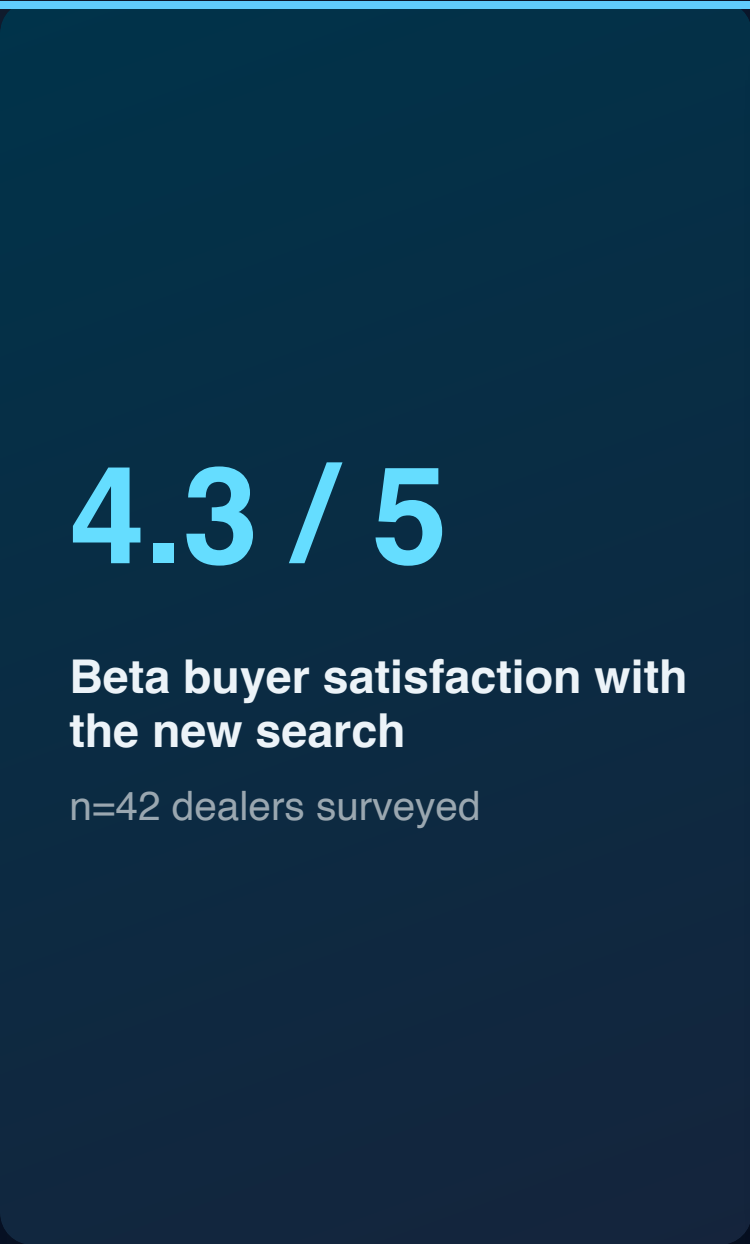
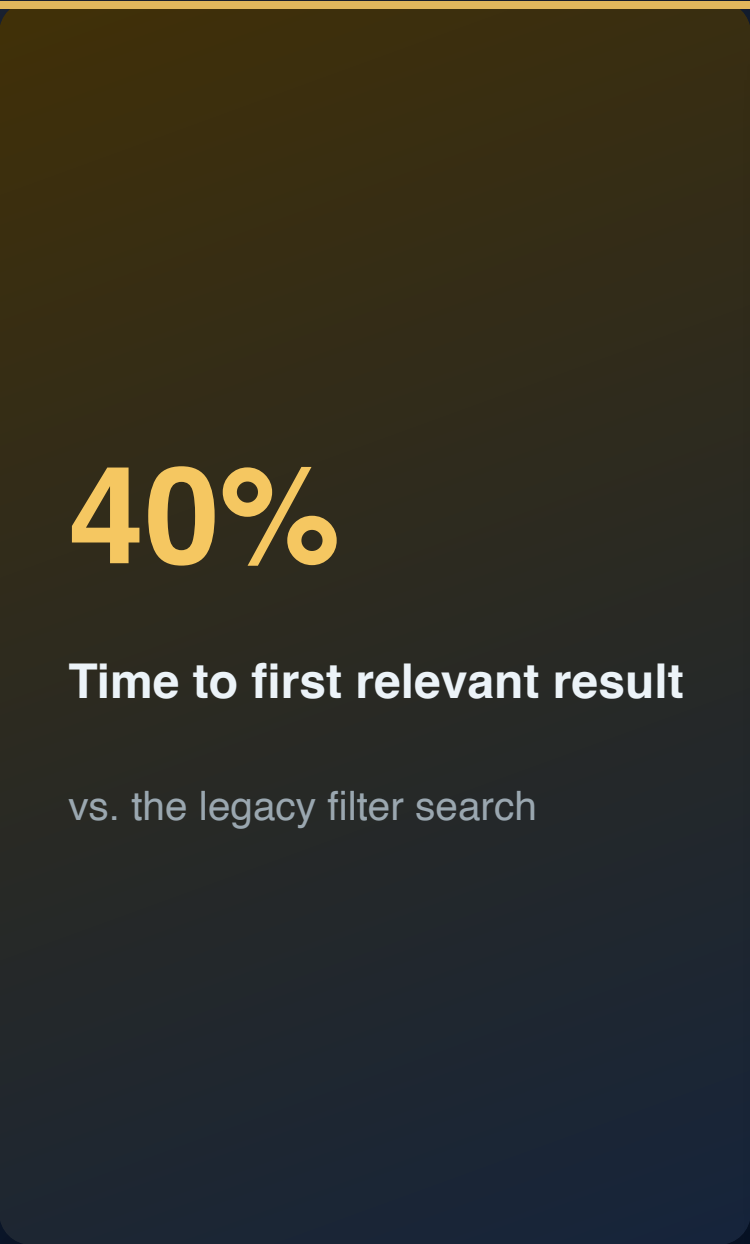
4Decide

IMPACT

# Early signals

*Illustrative results from beta — to be replaced with measured launch data.*

USER IMPACT



BUSINESS IMPACT

